Data Network Effects

More use → more data → better product

PROFESSIONAL DEFINITION

Data Network Effects describe a dynamic in which a product improves as more usage generates more data, which improves the product and attracts more users — a compounding, data-driven moat.

WHO DEVELOPED IT

Articulated by venture investors and platform scholars, notably Andrei Hagiu and Julian Wright, who cautioned that not all data improves products.

WHY IT WAS DEVELOPED

The concept was developed to distinguish genuine, defensible data advantages from the loose claim that 'more data wins', which often fails to compound.

FIGURE 1 · DATA NETWORK EFFECTS

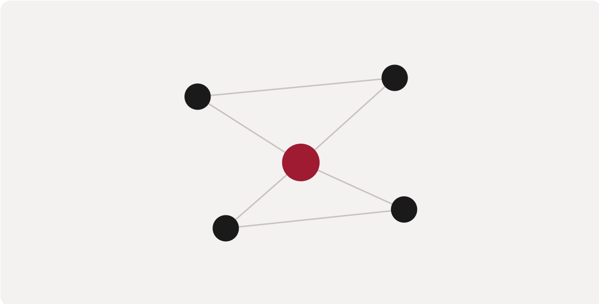

Figure 1. A schematic of the Data Network Effects as applied in BARBM312 (illustrative).

HOW IT IS USED

The team tests whether additional data actually improves the product and whether that improvement is hard for rivals to replicate; only then is a data moat real.

WHEN IT IS USED

When assessing AI product defensibility and investor due diligence on data-based advantage.

BY WHOM IT IS USED

Founders, product leaders and investors.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It refines the network-effects canon by isolating when data — not just users — creates compounding advantage, puncturing naïve 'big data' moats.

RELATED FRAMEWORKS IN THIS COURSE

- **Embedded vs. AI-Native 2×2** · Slibrary 3
- **Build · Buy · Partner** · Slibrary 3
- **Legacy Modernisation Curve** · Slibrary 3
- **Value Proposition Canvas** · Slibrary 3

SOURCES (HARVARD REFERENCING STYLE)

Hagiu, A. and Wright, J. (2020) 'When data creates competitive advantage', Harvard Business Review, 98(1).

Parker, G., Van Alstyne, M. and Choudary, S.P. (2016) Platform Revolution. New York: Norton.